RAJALAKSHMI ENGINEERING COLLEGE

ASSISTIVE TECHNOLOGY CENTRE OF EXCELLENCE

NIEPMD

SmartAbility

Innovation Challenge on Assistive Technology | Speech, Hearing & Communication

PROBLEM STATEMENT
PS-06

EchoSeed: Smart Articulation Training System

Team: Tech Knights **Members:** Spurgen A (Leader), Narean Raj D, Dharshini V, Suriya Narayanan S **Dept:** [Department] **Mentor:** [Mentor Name]

1 Problem Statement

Statement: Smart Articulation Training System. Speech sounds depend on precise coordination of the tongue, lips, jaw and airflow, but these movements are difficult for a child to observe while speaking.

Target users: Children with speech-sound (articulation) difficulties practising Hindi sounds, supported by speech-language pathologists (SLPs) and parents.

User persona: Aarav, age 6 (illustrative), says /t/ in place of /k/ (velar fronting). A mirror cannot show tongue placement, and home practice has little guidance until the next session.

2 Need & Gap

7.45%
of India's persons with disabilities have a speech disability (19,98,535). Census of India 2011

4.29%
of 2,304 rural school children aged 6–16 had communication disorders. Ballari study, Clin. Epidemiol. Glob. Health, 2021

4.41 per lakh
audiologists/SLPs per 1,00,000 people in India. RCI 2022, cited in Early Child Dev. Care, 2023

Existing solutions: Clinician instruction, mirror demonstration and imitation, with home practice between sessions

Limitations: A mirror shows only external movement; little immediate feedback between sessions; correct/incorrect results do not explain what to practise next

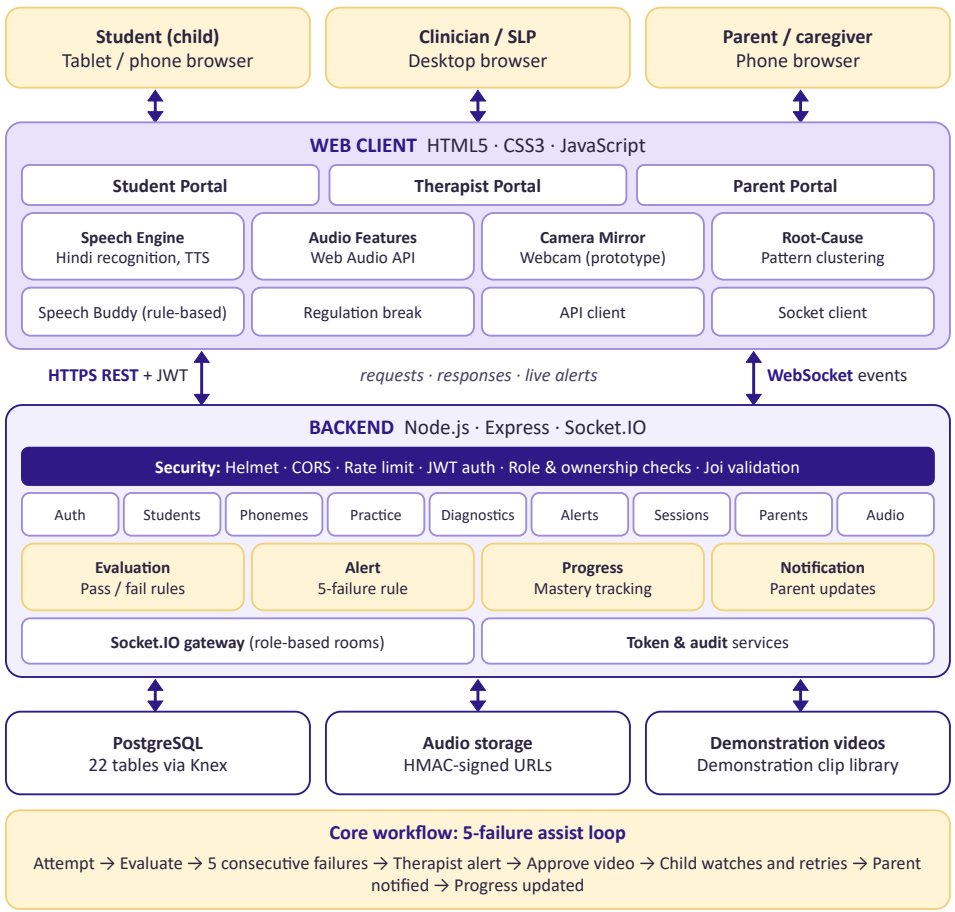
3 Proposed Solution

Concept: EchoSeed is a browser-based platform that helps children practise Hindi speech sounds with instant feedback, demonstration videos and clinician oversight.

Key Features

- Hindi speech recognition with slow audio (0.6x, 0.8x, 1.0x)
- Step-by-step levels: Sound → Word → Sentence
- Rule-based scoring with instant, child-friendly feedback
- 5 consecutive failures alert the therapist in real time
- Clinician approves or rejects each demonstration video
- Root-cause engine groups errors by shared articulatory feature
- Parent dashboard: mastery, practice heatmap and therapist notes

4 System Architecture



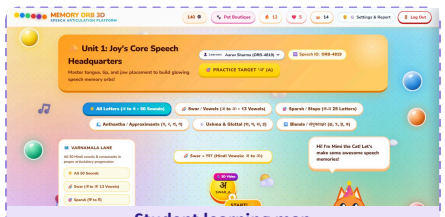
5 Methodology & Tech Stack

Research → Design → Develop → Deploy

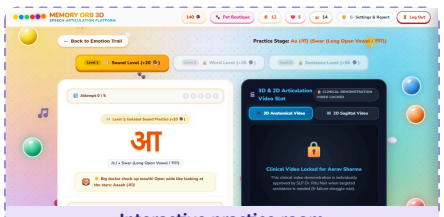
Workflow: Study articulation therapy needs, design the portals and 5-failure workflow, build the platform, launch the website

Tech stack: HTML5, CSS3, JavaScript, Web Speech API, Web Audio API, Node.js, Express, Socket.IO, PostgreSQL, Knex, JWT

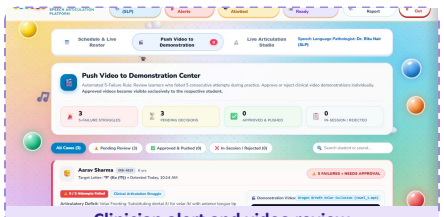
6 Prototype / Demo: Live Website

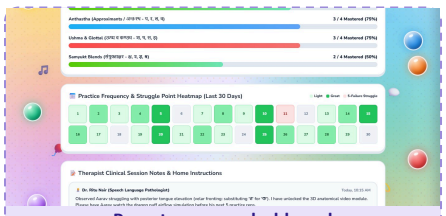


Student learning map



Interactive practice room


Clinician alert and video review


Parent progress dashboard

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Live website

7 Accessibility

- Web-based, with no installation needed
- Large, child-friendly tiles and gentle failure feedback
- Slow-cadence audio at 0.6x, 0.8x and 1.0x
- Dyslexia-friendly font and reduced-motion settings
- Calming regulation breaks during struggle

8 Impact & Scale

- Guided practice for children between sessions
- Helps SLPs spot struggling learners and error patterns
- Keeps parents informed with progress and notes
- Deployable via NIEPMD, schools and NGOs
- SDG 3, 4 & 10 alignment

9 Novelty & Future

- Explains error patterns, beyond correct/incorrect
- Clusters related sounds by shared articulatory cause
- Clinician stays in control of what the child sees
- Next: larger speech data and refined assessment
- Next: clinical validation with qualified SLPs

Acknowledgement
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Contact
[team email]

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Empowering Abilities • Driving Innovation • Building an Inclusive Future